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Data Center Energy Challenge: Can the US and Europe Deliver?

Artificial intelligence ambitions are colliding not just with the limits of semiconductors or algorithms. They face electricity and environmental bottlenecks.

By Maciej Bukowski and Jack Galloway

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It's a common European and American challenge: how to meet the rapid AI infrastructure buildout that is straining energy grids, raising energy bills, and generating environmental concerns. Without action, China could gain an unbeatable lead in rolling out AI.

US data centers already consume 4-5% of peak electricity demand, and estimates suggest that they will reach 12% by 2030. In Europe, data centers gobbled up almost 2.5% of the total energy supply in 2024, with electricity demand predicted to increase by 65% by 2030.

Specific countries already face a crunch. In Ireland, data centers already consume more than 22% of the country's electricity, stretching power grids. Irish household electricity bills are soaring, with the poorest households paying an estimated extra €209 between 2021 and 2023.

Governments face tough questions. How can they make sure data centers are built in the most efficient locations? Which types of energy sources should be considered green? And even if they manage to produce enough new electricity, will they be able to upgrade the grids fast enough to distribute it?

China holds an edge. It's permissive eminent domain laws allow for fast data center buildout. It coordinates energy planning and faces little resistance from its citizens. About 87% of Chinese citizens trust AI, and the government's coercive apparatus can crush dissent. From 2016 to 2023, the number of Chinese data center racks grew by 30% annually.

In contrast, data center construction in the US and Europe confronts hurdles, from lengthy environmental reviews, to securing high-end AI chips amid a global supply crunch and navigating local land use laws. In the US, state governments have created new cumbersome regulations. In Georgia, the Public Utilities Commission bills new energy customers that require more than 100 megawatts — essentially data centers — at higher rates. And in Ohio, a court decision will require data center customers to pay for 85% of predicted energy usage, regardless of actual consumption.

The upshot? Despite surging demand, data center capacity growth is slowing. In Europe, the Middle East, and Africa, it slumped in 2025 by 11% compared to 2024. In the US, almost 40% of data center projects have fallen behind schedule, as energy, equipment, and labor shortages threaten project progress.

A related debate concerns what type of energy to emphasize. Should nuclear be considered as green as wind or solar? France and Finland view nuclear generation as an important source of stable low-carbon electricity, crucial to supporting the data center buildout. But a staunch anti-nuclear coalition led by Germany and Austria insists renewable means renewable.The European Commission's new rating scheme

for data centers, which includes a sustainability metric, intentionally favors renewables over nuclear power. Although the Trump administration disdains renewables, US companies have committed millions of dollars to renewable sources.

Another challenge concerns where to locate data centers. Almost 7,000 out of 8,800 global data centers operate outside temperature ranges recommended for efficient operation, according to one analysis. Cold weather locations are more efficient than those in warm weather ones, because data centers must cool the enormous heat generated by their stacks of computers.

In the US, investment concentrates in states that offer rapid permitting processes, competitive electricity prices, favorable tax incentives, and available land. Southern states led by Texas, Arizona, Virginia, and Georgia account for 42% of all American data center construction.

In Europe, cloud operators have invested heavily in Nordic countries, with Microsoft alone investing €1.5 billion in 2024 and 2025. But many data centers need to be located near large population centers in central Europe. A YouTube video located in far-off Finland will load too slowly for users in northern Italy.

Water consumption has emerged as a significant concern. As frontier AI models demand chips with increasing computing capacity, traditional air-cooling techniques are proving inadequate. The alternative, water-based cooling, requires millions of

gallons of water per year, equivalent to the annual water consumption of three large hospitals.

Closed-loop cooling systems have become the norm in new data center construction projects. They mix water with glycol, a liquid coolant, to absorb the heat from the chips. Water is moved to a chiller plant, where large fans dissipate the heat. These systems require only an initial water input to the data center and consume annually the equivalent of just one small restaurant for water top-offs.

Translating these technological advances into on the ground reality remains a challenge. At present, just 15-20% of data centers are cooled using closed-loop systems.

But the potential revolution in water consumption showcases how innovation will be essential for the US and Europe to meet their ambitious AI infrastructure build-out plans. AI is often discussed as a software revolution. Increasingly, it resembles an infrastructure challenge. The availability of electricity, transmission capacity, cooling systems, and suitable locations may ultimately prove just as important as breakthroughs in machine learning.

Maciej Bukowski is the Head of Energy and Resilience Program at the Casimir Pulaski Foundation in Warsaw, and is a non-resident fellow with the Center for European Policy Analysis (CEPA), where he writes about issues including Central European security.

Jack Galloway is a Program Assistant with the Tech Policy Program at the Center for European Policy Analysis (CEPA). He holds a bachelor's degree in political science and environmental economics from the University of California, Berkeley.

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Das Google Datacenter, Kühltürme der Serverkühlung, Rechenzentrum in Eemshaven, Künstliche Intelligenz, auf über 44 Hektar, wird ausgebaut, betrieben mit Strom aus erneuerbaren Energien, Windkraft, Eemshaven, Niederlande, Google Datacenter *** The Google Datacenter, cooling towers for server cooling, data center in Eemshaven, artificial intelligence, on over 44 hectares, is being expanded, powered by electricity from renewable energies, wind power, Eemshaven, Netherlands, Google Datacenter No Use Switzerland. No Use Germany. No Use Austria